LESSON PLAN · SCRATCH (GR 6–8) ADVANCED: ALGORITHMS LAB

Lesson 10: Recursion: Blocks That Call Themselves

Module 4 | 45 minutes | Recursion | Grades 6–8 · ages 12–14

Objective: Write recursive procedures with a base case and a recursive case, and use recursion to draw a fractal tree.

Success criteria

- I can write a recursive procedure with a base case.
- I can draw a fractal with recursion.
- I can explain why the input must shrink.

Key vocabulary

- **recursion**: A procedure that calls itself.
- **base case**: The simple case where recursion stops.
- **recursive case**: The step that calls itself with a smaller input.
- **fractal**: A shape made of smaller copies of itself.

You will need

- Projector with Kitabash (lesson video, Code Lab)
- Devices: 1 per learner or pair
- Printed worksheet (optional)

Timed lesson plan (45 min)

Time	Stage	What happens
5 min	**Warm-up (unplugged)**	Russian dolls / paper folding (5 min): fold paper in half repeatedly — a natural base case appears.
5 min	**Watch**	Play the lesson video with captions; pause at each heading and ask a question.
8 min	**Build along**	Countdown and fractal tree: learners follow the steps.
14 min	**Perform**	Code Lab: recursive countdown, Code Lab: fractal tree
9 min	**Probe**	Debug the recursion, activity
4 min	**Wrap-up**	Exit ticket: What is a base case?

Unplugged warm-up

Russian dolls / paper folding (5 min): fold paper in half repeatedly — a natural base case appears.

Teaching the idea

Recursion: a procedure calls itself on a smaller sub-problem until a base case. Fractals demonstrate branching recursion; state must be restored (move back, turn back) so callers continue correctly. Missing base case → infinite recursion (stack overflow).

Analogy: Russian dolls: open one, find a smaller one, until the smallest.

Live demo: Trace countdown (3) as a stack of sticky notes on the board.

Questions to ask

Ask	Listen for
What is the base case in countdown?	n is 0 → Liftoff!.
Why must the input get smaller?	To reach the base case.
Why move back in branch?	To restore the position for the caller.

Differentiation

Support
Provide countdown; learners adapt it to count up.

Stretch
Draw a Koch snowflake or Sierpiński triangle recursively.

Common misconceptions

Misconception	How to fix it
Recursion is the same as a loop	It solves sub-problems by calling itself.
The base case is optional	Without it, recursion never ends.

Assessment

Use the Code Lab results (auto-checked), the quick checks and the Probe answers. Written answers arrive in your review queue.

Exit ticket
What is a base case?

Homework
Find three fractals in nature (ferns, broccoli, coastlines) and sketch one.

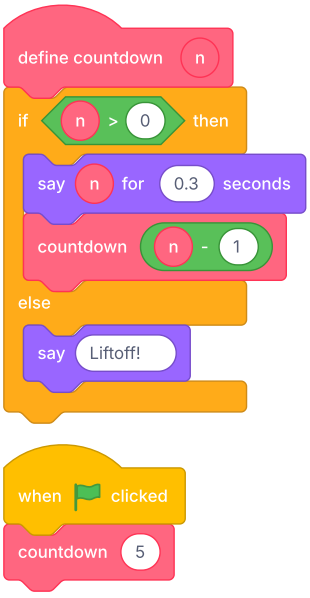
Answer key — Kitabash activities

1. Code Lab: recursive countdown (perform · code lab)

Goals checked: Counts 5, 4, 3, 2, 1, Liftoff! · Calls itself · Has a base case

One solution:

Coder

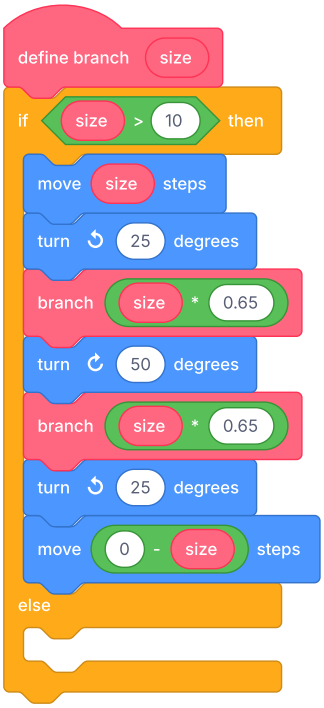


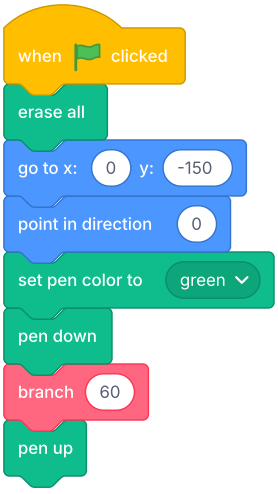
2. Code Lab: fractal tree (perform · code lab)

Goals checked: Recursive branch · A full tree · Smaller each time

One solution:

Pen





3. Debug the recursion (probe · multiple choice)

Answer: There is no base case to stop it

4. Explain your thinking (probe · free response)

Look for: base case, smaller, itself. **Example:** Recursion is when something is made of smaller copies of itself, like Russian dolls. To open them all, you open one doll, then do the same thing with the smaller doll inside. The base case is the tiny doll that doesn't open — that's when you stop.

Answer key — worksheet

1. **Key words:** recursion = A procedure that calls itself. · base case = The simple case where recursion stops. · recursive case = The step that calls itself with a smaller input. · fractal = A shape made of smaller copies of itself.
2. **Plan your code:** Goals: Counts 5, 4, 3, 2, 1, Liftoff!; Calls itself; Has a base case.
3. **Plan your code:** Goals: Recursive branch; A full tree; Smaller each time.
4. **Debug the recursion:** A — Without an if to stop at 0, it keeps calling itself forever.
5. **Explain your thinking:** Recursion is when something is made of smaller copies of itself, like Russian dolls. To open them all, you open one doll, then do the same thing with the smaller doll inside. The base case is the tiny doll that doesn’t open — that’s when you stop.

Kitabash original curriculum. Scratch is a project of the Scratch Foundation in collaboration with the MIT Media Lab; this course is not affiliated with or endorsed by them.